12TH CHEMISTRY · CHAPTER 32 · FULL CHAPTER NOTES

Medicine, Agriculture and Industry

Topic-wise short and long questions with answers from the complete chapter

PART I: MEDICINE — 32.1 Therapeutic index and window

Q.1 [Short] What is pharmacology? Why are safety limits of a drug important?

Pharmacology is the science which deals with what a drug does and the safety margins within which it operates. Every drug may be a potential poison; the difference lies in the dose. The therapeutic index and therapeutic window show how safe and effective a medicine is and help doctors to decide the right amount of a drug for a patient.

Q.2 [Long] What is the therapeutic window? Explain MEC, MTC and duration of action.

The therapeutic window is the range of drug concentrations in the plasma that are effective without causing harmful side effects. It is the “sweet spot” of medicine: not too low to be ineffective and not too high to be dangerous.

- **Minimum effective concentration (MEC):** the lowest dose below which the drug fails to produce a clinical effect.
- **Minimum toxic concentration (MTC):** the lowest dose at which the drug begins to cause adverse effects or “poisoning”.
- **Duration of action:** the time for which the drug concentration remains within its therapeutic window.

Q.3 [Short] Compare a wide and a narrow therapeutic window.

Wide therapeutic window	Narrow therapeutic window
Large gap between the dose that works and the dose that harms.	The effective dose is very close to the toxic dose.
The drug is safer; dosing is more flexible.	The drug is less safe; doctors must prescribe very carefully and often check the patient’s blood level.

The therapeutic window is also affected by the simultaneous use of several medicines, genetic factors, obesity, age and overall health.

Q.4 [Long] What is the therapeutic index? How is it calculated and interpreted?

The therapeutic index (TI) is a quantitative measurement of a drug’s safety. It compares the dose that causes a toxic effect with the dose that produces the desired therapeutic effect.

$$TI = TD_{50} \div ED_{50}$$

- **TD_{50} (Toxic Dose 50):** the dose that causes a toxic effect in 50% of the people who take it.

- **ED_{50} (Effective Dose 50):** the dose that works for 50% of the people who take it.

A higher TI means the drug is safer, because there is a larger difference between the effective and toxic doses; a drug with TI 10 is much safer than one with TI 2. A low TI means the drug is more dangerous and needs careful monitoring.

Figure 32.1 of the textbook prints "Therapeutic index = ED_{50}/TD_{50}". This is inverted: the text itself says TI compares the toxic dose to the effective dose and that a higher TI is safer, which requires TI = $TD_{50} \div ED_{50}$.

Q.5 [Short] Give examples of drugs with high and low therapeutic index.

High TI: penicillin and paracetamol; accidentally doubling the dose is not a threat to life. **Low TI:** warfarin (a blood thinner) and digoxin (a heart medicine); they must be taken only in the proper dose, and patients need regular blood tests called therapeutic drug monitoring (TDM). Digoxin toxicity causes ventricular fibrillation (a fatal heart rhythm) and yellow-tinted vision (xanthopsia).

Q.6 [Short] How does the therapeutic window differ from the therapeutic index?

Therapeutic window	Therapeutic index
A range of plasma concentrations (between MEC and MTC).	A ratio (number): $TD_{50} \div ED_{50}$.
Shows the concentrations at which the drug is effective without harm; guides dosing.	A quantitative measure of the safety of the drug.
Wide window = safer drug.	High TI = safer drug.

32.2 IMPORTANCE OF pH IN STOMACH

Q.7 [Short] How is the pH of the stomach regulated? How does the stomach protect itself?

The stomach has a highly acidic environment with a low pH (1.5–3.5). The acidity is regulated by **parietal cells**, which actively pump hydrogen ions (H^+) into the stomach. The acid is important for digesting food and killing bacteria. The stomach protects itself with a thick layer of mucus and bicarbonate ions, which neutralise acid at the cell surface.

Q.8 [Short] What are non-specific reactions? Why are some medicines given a protective coating?

The low pH of the stomach can cause **non-specific reactions**, i.e. it can chemically change some drugs; some drugs break down in the acid before they can be absorbed into the bloodstream. For this reason some medicines are given in a protective coating that dissolves only in the less acidic small intestine.

Q.9 [Short] What is an active metabolite?

An active metabolite is the active form produced when the body chemically changes a drug after it is taken. A drug that is inactive (or less active) as given and is converted in the body into its active form is called a prodrug; e.g. codeine is converted in the body into morphine, which gives the pain relief.

The learning outcome mentions active metabolites, but the chapter text does not define the term; this answer is from general knowledge.

32.3 MECHANISM OF DRUG ACTION

Q.10 [Short] Explain the structure, mechanism of action and uses of aspirin.

Aspirin (2-ethanoyloxybenzoic acid, acetylsalicylic acid) has a benzene ring carrying a carboxylic acid group and an ester group. **Mechanism:** it blocks the COX (cyclooxygenase) enzyme, so the body cannot make the prostaglandins that cause pain, fever and inflammation. **Uses:** relief of pain, fever and inflammation; in low doses it reduces blood clotting.

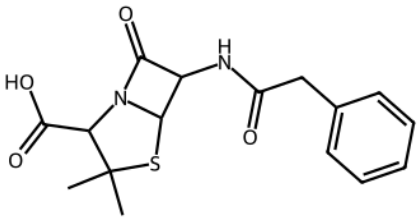
The textbook only says that the action of aspirin in blocking the COX enzyme was discussed in Chemistry XI (Chapter 3); the details and uses here are from that source and general knowledge.

Q.11 [Short] What is penicillin used for?

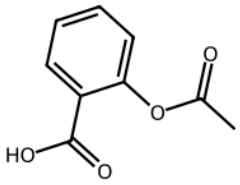
Penicillin is a powerful antibiotic used to treat many bacterial infections: respiratory infections such as strep throat, pneumonia, scarlet fever and sinusitis, and skin and soft-tissue infections such as cellulitis and erysipelas.

Q.12 [Long] Describe the structure and mechanism of action of penicillin.

Penicillin contains two core rings: a four-membered **beta-lactam ring** and a five-membered **thiazolidine ring**, both carrying side chains. The beta-lactam ring is crucial for the antibiotic activity; it is also the site where bacterial enzymes called beta-lactamases can attack and deactivate penicillin.



penicillin G



aspirin

Structures of a penicillin (penicillin G) and aspirin (own drawing)

1. The structure of penicillin resembles the amino acid sequence on which the bacterial enzyme that strengthens the cell wall (transpeptidase) normally acts.
2. The enzyme mistakenly takes up the penicillin molecule.
3. The beta-lactam ring opens and binds permanently to the enzyme, which is inactivated.

4. The bacteria continue to grow but without a functional cell wall, and eventually burst and die.
5. Human cells have no cell walls, so penicillin does not harm our cells.

Q.13 [Short] How was penicillin discovered?

In 1928 Alexander Fleming noticed that a mould growing on a contaminated petri dish had killed the surrounding bacteria. The mould, later identified as *Penicillium*, released a substance that inhibited bacterial growth. This chance observation led to the first antibiotic and revolutionised modern medicine.

Q.14 [Short] What are opiates? Give legal and illegal examples.

Opiates are a class of drugs that come from the opium poppy plant. They are powerful pain relievers but also cause addiction. Legal drugs include morphine, codeine, oxycodone and hydrocodone; illegal ones include heroin and fentanyl.

Q.15 [Long] Explain the concept of opioid receptors and their types.

The brain produces its own painkillers, called **endorphins**, which bind to opioid receptors in the central nervous system (CNS). Opiates "hijack" this natural system, which makes them effective for pain relief but also highly addictive.

Receptor	Location / role
Mu (μ)	In the brain and spinal cord; primary target of most opioids such as morphine. Responsible for analgesia, but activation also causes respiratory depression and intense euphoria.
Kappa (κ)	Involved in spinal analgesia and sedation; less associated with euphoria.
Delta (δ)	Play a vital role in emotional response and mood regulation.

Q.16 [Long] Describe the mechanism of action and uses of opiates.

1. Opiates bind to special proteins in the brain and spinal cord called opioid receptors, like a key (opiate) in a lock (receptor).
2. Binding closes the calcium channels and opens the potassium channels of the neuron.
3. This hyperpolarises the neuron, making it much harder for the cell to send the pain signal, so pain signals are blocked from reaching the brain.
4. Opiates also trigger the reward centre of the brain to release excess dopamine, giving a sense of relief and well-being (the cause of addiction).

Opioids relieve acute (sudden, short-term) pain and the severe pain of cancer. Codeine is used in cough syrups because it suppresses the cough reflex. Opioids also bind to receptors in the gastrointestinal tract, slow digestion and can relieve diarrhoea.

Q.17 [Short] Why did heroin cause a global addiction epidemic?

Heroin (diacetylmorphine) was once a trademarked medicine sold by Bayer as a "non-addictive" alternative to morphine and a cough suppressant for children. It is about five times as potent as morphine and crosses the blood–brain barrier faster, creating a "rush" that led to a global addiction epidemic.

32.4 CHALLENGES IN TREATING VIRAL INFECTIONS

Q.18 [Long] Why are viral infections difficult to treat with drugs?

1. **Viruses are parasites:** unlike bacteria, they are not truly alive on their own; they live inside our cells and use our own DNA/RNA machinery to replicate. It is hard to make a drug that kills the virus without harming healthy human cells.
2. **Mutation:** viruses change their genetic material very quickly, so a drug that works one year may not work the next; this is why flu vaccines are updated every year.
3. **Dormancy:** viruses can integrate into our genome and stay asleep for years; present drugs kill only active viruses.

Q.19 [Short] How do antiviral medications work?

Antiviral medications target specific stages of the viral life cycle: some prevent the virus from entering our cells; others stop it from copying its genetic material. A common mechanism is **enzyme inhibition**, e.g. protease inhibitors (used against HIV and hepatitis C). Some antivirals enhance the immune response: **interferons** are proteins that boost the immune system against viruses.

Quick Check 32.1

Q.20 [Short] a) Drug A has a TI of 3, and Drug B has a TI of 50. Which drug is considered safer, and why?

Drug B is safer. A higher therapeutic index means a larger difference between the effective dose and the toxic dose, so there is less risk of toxicity. Drug A (TI 3) has its toxic dose close to its effective dose and needs careful monitoring.

Q.21 [Short] b) How do antacids (e.g., magnesium hydroxide, $Mg(OH)_2$) chemically regulate stomach pH? Write a simple neutralization equation to illustrate this.

Antacids are weak bases. They neutralise the excess hydrochloric acid in the stomach, forming a salt and water, so the pH rises.

$$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$

Q.22 [Short] c) Identify the core, strained heterocyclic ring system in penicillin that is essential for its antibacterial activity.

The four-membered beta-lactam ring (fused to the five-membered thiazolidine ring).

Q.23 [Short] d) Compare the chemical structures of morphine and codeine (methylmorphine). How does this small structural difference relate to codeine's action as a prodrug?

Both have the same ring skeleton. Morphine has a free phenolic –OH group on its benzene ring; in codeine this group is methylated, i.e. it is an $–OCH_3$ (methyl ether) group. Codeine itself binds weakly to opioid receptors. In the body (liver) the methyl group is removed, converting part of the codeine into morphine, which is the active pain-relieving form. A drug that is changed into its active form inside the body is called a prodrug.

The structures of morphine and codeine are not given in the chapter; this answer is from general knowledge.

Q.24 [Short] e) Why do many antiviral drugs cause significant side effects for the host patient?

Viruses live inside our cells and use our own DNA/RNA machinery to replicate. A drug that stops the virus therefore also tends to interfere with the processes of healthy human cells, which causes side effects.

PART II: AGRICULTURE — 32.5 Fertilizers

Q.25 [Short] What are fertilizers? Differentiate synthetic and organic fertilizers.

Fertilizers are chemicals that provide essential nutrients to plants; they replenish depleted soil, improve its fertility, boost healthy growth and increase crop yields. **Synthetic fertilizers** are man-made by chemical processes such as the Haber-Bosch process for ammonia (NH_3). **Organic fertilizers** come from natural sources such as compost or animal manure.

Q.26 [Short] Which elements do plants need? What is the N-P-K ratio?

Plants require seventeen essential elements. **Micronutrients:** boron, iron, copper, chlorine, manganese, molybdenum and zinc. **Macronutrients:** carbon, hydrogen, oxygen, nitrogen, phosphorus, sulphur, potassium, calcium and magnesium. Nitrogen, phosphorus and potassium are needed in such large quantities that they are called **primary macronutrients**; commercial fertilizers carry an N-P-K ratio showing the concentration of these three.

Q.27 [Long] Describe the role of nitrogen, phosphorus and potassium in plant growth.

Nutrient	Function in the plant
Nitrogen (N)	Often the most limiting factor in crop production. A fundamental component of amino acids (which build proteins) and of chlorophyll, the pigment of photosynthesis.
Phosphorus (P)	Supplied as superphosphates or ammonium phosphates. A structural component of ATP, which stores and releases energy.
Potassium (K)	Does not become part of the plant's structure; acts as an osmoregulator and activates over sixty enzymes of plant metabolism.

Q.28 [Short] What is the impact of fertilizer application on soil health and water bodies?

Fertilizers are taken up by plant roots. If too much is used, the excess washes into rivers and lakes and causes an overgrowth of algae, which is harmful to aquatic life. Long-term use of some synthetic fertilizers can also change the soil's pH and kill beneficial microorganisms, hurting soil health. Crop rotation restores soil nutrients and reduces the need for synthetic fertilizers.

32.6 PESTICIDES

Q.29 [Long] What are pesticides? Describe their types and modes of action.

Pesticides are chemicals used to kill or control pests such as insects, weeds or fungi. They protect crops from damage, prevent food loss and ensure a stable food supply.

Type	Target	Mode of action / examples
Insecticides	Insects	Target the nervous system: organophosphates and neonicotinoids are neurotoxins causing paralysis and death. (Neonicotinoids are linked to Colony Collapse Disorder in honeybees.)
Herbicides	Weeds	Some inhibit enzymes so that essential amino acids are not made; some disrupt electron transport and stop photosynthesis.
Fungicides	Fungi	Target ergosterol synthesis in the fungal cell membrane; the cell leaks and dies. Examples: triazoles, strobilurins, zinc phosphide.

Q.30 [Long] What are the benefits and risks of using pesticides?

Benefits	Risks
Protect plants from pests and diseases and increase crop yield.	Can be carcinogenic and neurotoxic to humans.
Reduce the risk of epidemics and improve human health.	Harm beneficial insects, birds and non-target species such as pollinators (bees).
Contribute to food security and stability worldwide.	Run-off from fields contaminates water and harms aquatic organisms.
	Pests develop immunity (resistance), forcing the use of more toxic pesticides.

32.7 ACID RAIN

Q.31 [Short] What is acid rain? How is it formed?

Acid rain is an atmospheric mixture of nitric acid and sulphuric acid produced by industrial pollution. Normal rain water has a pH of about 5.6 or above (due to carbonic acid); rain with a pH lower than 5.6 is acid rain. It is caused by sulphur oxides (SO_x) and nitrogen oxides (NO_x), which react with water to form H_2SO_4 and HNO_3.

$$2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4 \text{ (sulphuric acid formation)}$$
$$4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3 \text{ (nitric acid formation)}$$

Q.32 [Long] Which reactions occur when acid rain falls on soil? How do they affect crops?

Acid rain leaches essential nutrients such as Ca^{2+}, Mg^{2+} and K^+ out of the soil, making them unavailable to plants. It also increases the solubility of toxic metals such as Al^{3+}, which are absorbed by the roots and damage them. The plant takes up fewer nutrients, so growth is slower and crop yields are lower.

$$K^+\text{—Soil} + H^+ \rightarrow H^+\text{—Soil} + K^+ \text{ (potassium leaching)}$$
$$Ca^{2+}\text{—Soil} + 2H^+ \rightarrow H_2{}^+\text{—Soil} + Ca^{2+} \text{ (calcium leaching)}$$
$$Al^{3+}\text{—Soil} + 3H^+ \rightarrow H_3{}^+\text{—Soil} + Al^{3+} \text{ (aluminium leaching)}$$

- Direct acid rain damages and dehydrates the leaves and makes the plant vulnerable to disease.
- It reduces soyabean and wheat yields by reducing nitrogen fixation and nutrient intake.
- It harms soil microbial communities and disrupts natural nutrient cycles and the decomposition of organic matter.
- In lakes, the Al^{3+} released from soil clogs fish gills and reduces oxygen absorption, so fish suffocate.

32.8 EFFECT OF CLIMATE CHANGE ON CROPS

Q.33 [Long] How do changes in temperature, precipitation and extreme weather affect crops?

Change	Effect on crops
Temperature	Higher temperature speeds up metabolism, but heat stress causes dehydration, poor photosynthesis and respiration, and lower growth and yield. Cold damages certain crops and is fatal to young plants.
Precipitation	Normal rain supplies water and nutrients. Changed rainfall patterns cause drought in some areas and floods in others, both causing crop failure. Unpredictable rains and cloud bursts erode the nutrient-rich top soil.
Extreme weather	Hurricanes, heatwaves and hailstorms destroy crops and infrastructure, causing food shortages. Storms uproot plants; strong winds cause water loss; hail damages leaves, fruit and stems; heat waves and high UV cause water loss and leaf burning.

Q.34 [Short] How are scientists making crops resilient to climate change?

To adapt, scientists are developing crop varieties that are more resilient to changing climates, e.g. drought-resistant corn, heat-tolerant wheat and flood-proof “Sub1 rice”, which can survive being submerged under water for two weeks.

32.9 GENETIC ENGINEERING

Q.35 [Long] What is genetic engineering? How is a GM crop developed?

Genetic engineering in agriculture involves changing an organism’s DNA by adding, removing or modifying specific genes. It is used to create genetically modified (GM) crops with desired traits.

1. **Identification:** scientists identify a useful gene with a desired trait (e.g. pest resistance from a bacterium, or drought tolerance).
2. **Isolation:** the gene responsible for the trait is isolated from the organism.
3. **Insertion:** the gene is inserted into the target plant’s genome using a gene gun or a carrier bacterium.
4. **Expression:** the plant’s cells express the new gene; the plant now produces a new protein and shows the new trait.

Q.36 [Short] What are the benefits and risks of GM crops?

Benefits	Risks
Improved resistance to pests and diseases.	GM genes may spread to wild plants (outcrossing).
Tolerance to herbicides.	“Superweeds”: herbicide-resistance genes jump to wild weeds by cross-pollination.
Enhanced nutritional value, e.g. Golden Rice with beta-carotene to prevent blindness.	Resistant insects are produced.
Higher yields and less need for pesticides, e.g. BT-cotton makes its own pesticide.	Unknown long-term effects on human health and biodiversity.

Quick Check 32.2

Q.37 [Short] a) State two major agricultural benefits of using fungicides, linking to crop health or yield.

(1) They kill fungi that cause plant diseases, keeping the crop healthy and reducing the risk of epidemics. (2) By preventing damage and food loss they increase crop yield and contribute to a stable food supply.

Q.38 [Short] b) Write the balanced chemical equation for reaction that occurs when powdered limestone ($CaCO_3$) is added to a lake or field to neutralize sulphuric acid (H_2SO_4) from acid rain.

$$CaCO_{3(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + H_2O_{(l)} + CO_{2(g)}$$

Q.39 [Short] c) Identify two specific traits (e.g., heat tolerance, salt tolerance) that scientists aim to develop in new crop varieties to make them more resilient to climate change.

(1) Drought resistance (e.g. drought-resistant corn). (2) Flood tolerance (e.g. Sub1 rice, which survives two weeks under water). Heat tolerance (heat-tolerant wheat) is another.

PART III: INDUSTRY — 32.10 Raw materials in Pakistan

Q.40 [Short] Justify the importance of industrial chemistry in different sectors.

Industrial chemistry is crucial for the economy and daily life. It serves manufacturing (plastics, fibres, cement, steel, glass, paper, detergents), energy (refining crude oil and natural gas into fuels), healthcare (medicines made by the pharmaceutical industry) and environmental protection (water-treatment chemicals such as chlorine, and safety and pollution control in plants).

The chapter introduces these sectors in one line; the examples are collected from sections 32.10–32.14.

Q.41 [Long] Which raw materials are used in industrial chemistry? Which are available in Pakistan?

The feedstock of an industry determines its location and economic viability. Air provides nitrogen for fertilizers and oxygen for steel making; water is a solvent, a cooling agent and a source of hydrogen; fossil fuels provide the hydrocarbons for 90% of organic chemicals. Raw materials available in Pakistan:

Raw material	Use
Natural gas	Major source of methane (CH_4); used to produce fertilizers (urea) and petrochemicals; both a fuel and a source of hydrogen.
Rock salt (Khewra mines)	One of the world's largest deposits; source of NaCl for soda ash (Na_2CO_3), caustic soda (NaOH) and chlorine.
Limestone	Abundant in the Salt Range and Sindh; source of $CaCO_3$ for the cement industry.
Gypsum	Cement industry and plaster; source of calcium and sulphur; soil conditioner.
Chromite	Found in Muslim Bagh; used in stainless steel and pigment production.
Crude oil	Refined to produce fuels and petrochemicals.
Sulphur	Used to make sulphuric acid (H_2SO_4).

32.11 CHEMICAL INDUSTRIES AND THE ECONOMY

Q.42 [Long] Discuss the importance of chemical industries in the economy of Pakistan.

- The chemical industry contributes significantly to the Gross Domestic Product (GDP) and provides jobs.
- It supports agriculture (fertilizers), construction (cement) and textiles (dyes and synthetic fibres).
- It uses locally available raw materials (natural gas, rock salt, limestone, crude oil), which reduces reliance on imports.
- Food security: local companies use natural gas to produce urea, cutting the import bill by billions of dollars and making fertilizer affordable.
- Textiles (over 50% of exports) consume caustic soda for mercerizing cotton, synthetic dyes for colour and hydrogen peroxide for bleaching.
- The cement industry has become a leading exporter to neighbouring regions.

32.12 INDUSTRIAL CHEMICAL PROCESSES

Q.43 [Long] **Describe the main industrial chemical processes, with Pakistani examples.**

1. Polymerisation: small molecules (monomers) link to form large chains (polymers). In *addition polymerisation* monomers add to one another without losing any atoms (ethene → polyethene). In *condensation polymerisation* monomers join and release a small molecule, usually water (polyesters). Engro Polymer and Chemicals is the sole producer of PVC in Pakistan.

2. Refining and reforming: fractional distillation separates crude oil into gasoline, diesel, kerosene etc. by boiling point. Reforming converts straight-chain hydrocarbons into branched ones over a heated catalyst to give high-octane gasoline. Refineries such as PARCO and Attock Refinery supply petrochemical feedstocks.

3. Haber-Bosch process: ammonia is synthesised from nitrogen and hydrogen with iron as catalyst. It accounts for 13% of the nitrogen fixation on earth; 70% of ammonia goes into fertilizers. Engro Fertilizers and FFC use natural gas as feedstock and convert the ammonia into urea.

$$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} \text{ (Fe catalyst)}$$

4. Contact process: SO_2 is converted into SO_3 using V_2O_5 or Pt as catalyst; SO_3 is changed into oleum and finally into highly pure, concentrated H_2SO_4, the "King of Chemicals". Example: Nimir Industrial Chemicals.

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)} \ (V_2O_5)$$

5. Solvay process: produces sodium carbonate (Na_2CO_3) from brine and limestone. Ammonia is added to brine and CO_2 is passed to precipitate $NaHCO_3$, which on heating gives Na_2CO_3. Ammonia is regenerated using lime; the by-product is $CaCl_2$. Lucky Core Industries (formerly ICI) makes soda ash for glass, detergents and paper.

6. Chlor-alkali process: electrolysis of brine (NaCl) gives chlorine (Cl_2), sodium hydroxide (NaOH) and hydrogen (H_2), using modern membrane cells. The products are used for plastics, detergents and water treatment. Sitara Chemical Industries and Ittehad Chemicals are the primary producers.

$$2NaCl_{(aq)} + 2H_2O_{(l)} \rightarrow 2NaOH_{(aq)} + Cl_{2(g)} + H_{2(g)} \text{ (electrolysis)}$$

32.13 APPLICATIONS OF INDUSTRIAL CHEMISTRY

Q.44 [Long] Explain the applications of industrial chemistry in different industries.

Industry	Role of industrial chemistry
Petrochemical	Crude oil is cracked into smaller molecules, the building blocks of plastics, synthetic rubber, fibres, lubricants and waxes.
Cosmetics	Makes ingredients for makeup, soaps and perfumes: surfactants (soaps), emulsifiers (smooth creams) and preservatives (prevent bacterial growth); products are pH-balanced for skin.
Cement	An intimate mixture of limestone and clay is heated to 1500 °C in a rotary kiln; the product is ground with gypsum. (Cement warms with water: exothermic hydration.)
Food	Additives and preservatives (sodium benzoate); hydrogenation of edible oils to solid fats like margarine; leavening agents (sodium bicarbonate); fortification (iodine in salt, vitamin A in oil).

32.14 SAFETY MEASURES AND PRECAUTIONS

Q.45 [Long] What safety measures are necessary in industrial chemical facilities?

1. **Personal protective equipment (PPE):** gloves, goggles, helmets, lab coats and protective shoes to avoid injuries from chemicals, heat or harmful fumes.
2. **Labelling, handling and storage:** chemicals must be properly labelled, handled and stored according to their properties; flammable, toxic or corrosive substances are kept in suitable containers and separate storage areas to prevent leaks, spills, contamination or dangerous reactions.
3. **Emergency plans:** clear plans for fires, chemical spills, gas leaks or explosions; emergency exits, fire extinguishers, first-aid kits and spill-control materials must be available and workers must know how to respond.
4. **Ventilation:** good ventilation systems remove toxic gases, fumes and dust and keep the workplace safe and healthy.
5. **Training and maintenance:** workers must receive safety training on hazardous materials and equipment; regular inspection and maintenance of machines, pipelines and storage tanks detects faults early.

Quick Check 32.3

Q.46 [Short] a) Identify two major sectors of Pakistan's economy (other than agriculture) that are heavily reliant on the chemical industry, and name one chemical product essential for each.

(1) Textiles: caustic soda for mercerizing cotton (also synthetic dyes and hydrogen peroxide for bleaching). (2) Construction: cement.

Q.47 [Short] **b) Explain the physical principle of the process used to separate crude oil (in terms of intermolecular forces and boiling points) on which this separation depends.**

Crude oil is separated by fractional distillation, which depends on differences in boiling point. Larger hydrocarbon molecules have stronger intermolecular (London dispersion) forces, so they have higher boiling points and condense lower in the column; smaller molecules have weaker forces and lower boiling points and rise higher before condensing.

Q.48 [Short] **c) Many cosmetics and perfumes use esters. Name the two types of reactants needed to synthesize an ester and name the catalyst typically used.**

A carboxylic acid and an alcohol; the catalyst is concentrated sulphuric acid.

$$RCOOH + R'OH \rightleftharpoons RCOOR' + H_2O \text{ (conc. } H_2SO_4)$$

Q.49 [Short] **d) Explain the chemical reason why highly oxidizing chemicals (like concentrated HNO_3) must be stored separately from flammable organic chemicals (like ethanol).**

A strong oxidising agent readily oxidises organic compounds, which act as fuels (reducing agents). If they come into contact, a fast, highly exothermic redox reaction can occur, which may cause a fire or an explosion.

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